David Talmy
Associate Professor
Department of Microbiology, University of Tennessee
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Education

2009-2013 **PhD Environmental Science**
School of Biological Sciences, University of Essex

2008-2009 **MRes Mathematics in the Living Environment (with Distinction)**
Departments of Biology and Mathematics, University of York

2004-2008 **MMath Mathematics**
School of Mathematical and Physical Sciences, University of Sussex

Professional Experience

2025-present **Associate Professor**
Department of Microbiology and the National Institute of Mathematical and Biological Synthesis (NIMBioS), University of Tennessee

2025-present **Adjunct Associate Professor**
Department of Ecology and Evolutionary Biology, University of Tennessee

2018-2025 **Assistant Professor**
Department of Microbiology and the National Institute of Mathematical and Biological Synthesis (NIMBioS), University of Tennessee

2020-2025 **Adjunct Assistant Professor**
Department of Ecology and Evolutionary Biology, University of Tennessee

2016-2018 **Research Scientist**
Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology

2013-2016 **Postdoctoral Associate**
Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology

2010-2013 **Graduate Research Assistant**
Plymouth Marine Laboratory

Research Interests

Global marine ecosystem structure and function, marine biogeochemistry, predator-prey interactions, microbial physiology and metabolism.

Peer-Reviewed Publications

Underlined names indicate trainees in my research group

1. Hambrick, K.M., Smith, L.E., Martin, R.M., Wei, B., Dey, R., Boyer, G.L., Weitz, J.S., **Talmy, D.**, Wilhelm, S.W., Zinser, E.R. (2026). Acclimation temperature influences phage susceptibility in a toxin-producing strain of *Microcystis aeruginosa*. *Microbiology Spectrum*, 14(7), e03379-25.

2. Dey, R., Beach, R., Hambrick, K.M., Sgouralis, I., Frémont, P., Demory, D., Carr, E., Beckett, S.J., Weitz, J.S., **Talmy, D.** (2026). Microbial Primer: Bayesian learning of traits from microbial time series data. *Microbiology*, 172(8).

3. McCullough, D.K., **Talmy, D.** (2026). Resource competition modeling suggests hydrogen peroxide determines competitive outcomes among oligotrophic cyanobacteria. *ISME Communications*, 6(1), ycag188.

4. Frémont, P., Beckett, S.J., Demory, D., Carr, E., Follett, C.L., Lindell, D., **Talmy, D.**, Dutkiewicz, S., Weitz, J.S. (2026). Coexistence of photosynthetic marine microorganisms, viruses and grazers: towards integration in ocean ecosystem models. *Environmental Microbiology*, 28(4), e70295.

5. McCullough, D.K., Bowden, E.C., Calfee, B.C., Gilchrist, M.A., Zinser, E.R., **Talmy, D.** (2026). Quantification of the coupled dynamics of marine microbes and reactive oxygen species in laboratory batch culture experiments. *Microbiology Spectrum*, 14(6), e00712-25.

6. **Talmy, D.**, Howard-Varona, C., Eveillard, D., Covert, M., Sullivan, M.B. (2025). Viruses in multi-scale ocean models: challenges and opportunities. *Frontiers in Marine Science*, 12, 1717845.

7. McCoy, S., McBride, D., McCullough, D.K., Calfee, B.C., Zinser, E.R., **Talmy, D.**, Sgouralis, I. (2025). Quantitative assessment of biological dynamics with aggregate data. *Bulletin of Mathematical Biology*, 87(11), 164.

8. Martínez Martínez, J., **Talmy, D.**, Kimbrel, J.A., Weber, P.K., Mayali, X. (2024). Coastal bacteria and protists assimilate viral carbon and nitrogen. *The ISME Journal*, 18(1), wrae231.

9. Lennon, J., Abramoff, R., Allison, S., Burckhardt, R., DeAngelis, K., Dunne, J., Frey, S., Friedlingstein, P., Hawkes, C., Hungate, B., Khurana, S., Kivlin, S., Levine, N., Manzoni, S., Martiny, A., Martiny, J., Nguyen, N., Rawat, M., **Talmy, D.**, Todd-Brown, K., Vogt, M., Wieder, W., Zakem, E. (2024). Priorities, opportunities, and challenges for integrating microorganisms into Earth system models for climate change prediction. *mBio*, 15(5), e00455-24.

10. **Talmy, D.**, Carr, E., Rajakaruna, H., Våge, S., Omta, A.W. (2024). Killing the predator: impacts of highest-predator mortality on the global-ocean ecosystem structure. *Biogeosciences*, 21(10), 2493-2507.

11. Gross, L.J., McCord, R.P., LoRe, S., Ganusov, V.V., Hong, T., Strickland, C., **Talmy, D.**, von Arnim, A.G., Wiggins, G. (2023). Prioritization of the concepts and skills in quantitative education for graduate students in biomedical science. *PLOS ONE*, 18(4), e0284982.

12. Omta, A.W., Heiny, E.A., Rajakaruna, H., **Talmy, D.**, Follows, M.J. (2023). Trophic model closure influences ecosystem response to enrichment. *Ecological Modelling*, 475, 110183.

13. Hinson, A., Papoulis, S., Fiet, L., Knight, M., Cho, P., Szeltner, B., Sgouralis, I., **Talmy, D.** (2023). A model of algal-virus population dynamics reveals underlying controls on material transfer. *Limnology and Oceanography*, 68(1), 165-180.

14. Rajakaruna, H., Carr, E., Omta, A.W., **Talmy, D.** (2023). Linear scaling between microbial predator and prey densities in the global ocean. *Environmental Microbiology*, 25(2), 306-314.

15. **Talmy, D.** (2022). Life, death and cyanobacterial biogeography. *Nature Microbiology* (News and Views), 7(4), 480-481.

16. Garcia, N.S., **Talmy, D.**, Fu, W., Larkin, A.A., Lee, J., Martiny, A.C. (2022). The diel cycle of surface ocean elemental stoichiometry has implications for ocean productivity. *Global Biogeochemical Cycles*, 36(3), e2021GB007092.

17. Martiny, A.C., Hagstrom, G.I., DeVries, T., Letscher, R.T., Britten, G.L., Garcia, C.A., Galbraith, E., Karl, D., Levin, S.A., Lomas, M.W., Moreno, A.R., **Talmy, D.**, Wang, W., Matsumoto, K. (2022). Marine phytoplankton resilience may moderate oligotrophic ecosystem responses and biogeochemical feedbacks to climate change. *Limnology and Oceanography*, 67(S1), S378-S389.

18. Cael, B.B., Bisson, K., Conte, M., Duret, M.T., Follett, C.L., Henson, S.A., Honda, M.C., Iversen, M.H., Karl, D.M., Lampitt, R.S., Mouw, C.B., Muller-Karger, F., Pebody, C.A., Smith, K.L., **Talmy, D.** (2021). Open ocean particle flux variability from surface to seafloor. *Geophysical Research Letters*, 48(9), e2021GL092895.

19. Papoulis, S.E., Wilhelm, S.W., **Talmy, D.**[*], Zinser, E.R.[*] (2021). Nutrient loading and viral memory drive accumulation of restriction modification systems in bloom-forming cyanobacteria. *mBio*, 12(3), e00873-21.
[*]*co-corresponding authors*.

20. Omta, A.W., **Talmy, D.**, Inomura, K., Irwin, A.J., Finkel, Z.V., Sher, D., Liefer, J.D., Follows, M.J. (2020). Quantifying nutrient throughput and DOM production by algae in continuous culture. *Journal of Theoretical Biology*, 494, 110214.

21. Pound, H.L., Gann, E.R., Tang, X., Krausfeldt, L.E., Huff, M., Staton, M.E., **Talmy, D.**, Wilhelm, S.W. (2020). The "neglected viruses" of Taihu: abundant transcripts for viruses infecting eukaryotes and their potential role in phytoplankton succession. *Frontiers in Microbiology*, 11, 338.

22. Inomura, K., Omta, A.W., **Talmy, D.**, Bragg, J., Deutsch, C., Follows, M.J. (2020). A mechanistic model of macromolecular allocation, elemental stoichiometry, and growth rate in phytoplankton. *Frontiers in Microbiology*, 11, 86.

23. **Talmy, D.**, Beckett, S.J., Taniguchi, D.A.A., Brussaard, C.P.D., Weitz, J.S., Follows, M.J. (2019). An empirical model of carbon flow through marine viruses and microzooplankton grazers. *Environmental Microbiology*, 21(6), 2171-2181.

24. Nissimov, J.I., **Talmy, D.**, Haramaty, L., Fredricks, H.F., Zelzion, E., Knowles, B., Eren, A.M., Vandzura, R., Laber, C.P., Schieler, B.M., Johns, C.T., More, K.D., Coolen, M.J.L., Follows, M.J., Bhattacharya, D., Van Mooy, B.A.S., Bidle, K.D. (2019). Biochemical diversity of glycosphingolipid biosynthesis as a driver of *Coccolithovirus* competitive ecology. *Environmental Microbiology*, 21(6), 2182-2197.

25. **Talmy, D.**, Beckett, S.J., Zhang, A.B., Taniguchi, D.A.A., Weitz, J.S., Follows, M.J. (2019). Contrasting controls on microzooplankton grazing and viral infection of microbial prey. *Frontiers in Marine Science*, 6, 182.

26. Thamatrakoln, K., **Talmy, D.**, Haramaty, L., Maniscalco, C., Latham, J.R., Knowles, B., Natale, F., Coolen, M.J.L., Follows, M.J., Bidle, K.D. (2019). Light regulation of coccolithophore host–virus interactions. *New Phytologist*, 221(3), 1289-1302.

27. Omta, A.W., **Talmy, D.**, Sher, D., Finkel, Z.V., Irwin, A.J., Follows, M.J. (2017). Extracting phytoplankton physiological traits from batch and chemostat culture data. *Limnology and Oceanography: Methods*, 15(5), 453-466.

28. Li, G., **Talmy, D.**, Campbell, D.A. (2017). Diatom growth responses to photoperiod and light are predictable from diel reductant generation. *Journal of Phycology*, 53(1), 95-107.

29. Record, N.R., **Talmy, D.**, Våge, S. (2016). Quantifying tradeoffs for marine viruses. *Frontiers in Marine Science*, 3, 251.

30. Lopez, J.S., Garcia, N.S., **Talmy, D.**, Martiny, A.C. (2016). Diel variability in the elemental composition of the marine cyanobacterium *Synechococcus*. *Journal of Plankton Research*, 38(4), 1052-1061.

31. **Talmy, D.**, Martiny, A.C., Hill, C.N., Hickman, A.E., Follows, M.J. (2016). Microzooplankton regulation of surface ocean POC:PON ratios. *Global Biogeochemical Cycles*, 30(2), 311-332.

32. **Talmy, D.**, Blackford, J., Hardman-Mountford, N.J., Polimene, L., Follows, M.J., Geider, R.J. (2014). Flexible C:N ratio enhances metabolism of large phytoplankton when resource supply is intermittent. *Biogeosciences*, 11(17), 4881-4895.

33. **Talmy, D.**, Blackford, J., Hardman-Mountford, N.J., Dumbrell, A.J., Geider, R.J. (2013). An optimality model of photoadaptation in contrasting aquatic light regimes. *Limnology and Oceanography*, 58(5), 1802-1818.

Preprints

34. **Talmy, D.**, Carr, E., Frémont, P., Demory, D., Follett, C.L., Jahn, O., Muratore, D., Beckett, S.J., Lindell, D., Weitz, J.S., Dutkiewicz, S. (2026). Implicit representation of viruses fails to capture impacts of virus-induced mortality in global marine ecosystem models. *bioRxiv*. https://doi.org/10.64898/2026.07.21.739416

35. Frémont, P., Beckett, S.J., Muratore, D., Demory, D., Carr, E., Jahn, O., Follett, C.L., **Talmy, D.**, Lindell, D., Weitz, J.S., Dutkiewicz, S. (2026). Viral impacts on plankton standing stocks, primary productivity, and biogeochemistry in a model ocean. *bioRxiv*. https://doi.org/10.64898/2026.05.27.728272

Non-Peer-Reviewed Publications

1. Microbes in Models: Integrating microbes into Earth System Models for understanding climate change: Report on an American Academy of Microbiology virtual colloquium held on Dec. 6 and 8, 2022. Washington (DC): American Society for Microbiology, 2023.

Grants and Awards

1/1/2027-12/31/2029	NSF co-Principal Investigator award (Total award: $1,454,878; $245,469 to co-PI David Talmy) "*How temporal niche partitioning and growth on urea selects for cyanobacteria in fresh waters*". ID number: DEB-2534721

5/1/2025- 4/30/2028	NSF Principal Investigator award (Total award: $1,607,111; $454,006 to PI David Talmy) "*Collaborative Research: Constraining virus loss and carbon and nitrogen flow through virovory in temperate marine ecosystems with isotope tracing and microbial food web models*". ID number: OCE-2445508
10/1/2024- 9/30/2027	Simons Foundation Principal Investigator award (Total award: $226,805; full amount to PI David Talmy) "*SCOPE Synthesis: vDarwin Initiative*". ID number: SFI-LS-Project-00010539
4/15/2024- 3/31/2029	NSF and NIEHS co-Principal Investigator award (Total award: $7,382,485 to University of Michigan; $573,606 subaward to UTK; $160,001 to co-PI David Talmy) "*Great Lakes center for fresh waters and human health*". ID number: 2418066
8/1/2020- 7/31/2025	NSF Principal Investigator award (Total award: $913,189; $397,804 to PI David Talmy, $515,385 to co-PI Erik Zinser) "*Characterizing the effects of exogenous reactive oxygen species on marine microbial ecosystem dynamics*". ID number: OCE-2023680
4/1/2020- 3/31/2023	Simons Early Career Investigator in Marine Microbial Ecology and Evolution Award (Total award: $666,000; full amount to PI David Talmy). ID number: 690671 (NCE until 3/31/2024)
10/1/2018- 9/30/2020	Burroughs Wellcome Fund co-Principal Investigator award (Total award: $150,000 to PI Louis Gross; $18,750 to co-PI David Talmy) "*Enhancing Quantitative and Data Science Education for Graduate Students in Biomedical Science at the University of Tennessee, Knoxville*". ID number: 1018963
6/1/2018- 9/30/2019	NSF Principal Investigator award (Total award: $55,642 to PI David Talmy; transfer of unspent funds from OCE-1536521 from MIT to UTK) "*Elucidating algal host-virus dynamics in different nutrient regimes - mechanistic interactions and biogeochemical impact*". ID number: OCE-1849926
9/1/2015- 10/31/2019	NSF co-Principal Investigator award (Total award: $326,864 to PI Mick Follows; co-PI David Talmy was a postdoc on the award) "*Elucidating algal host-virus dynamics in different nutrient regimes - mechanistic interactions and biogeochemical impact*". ID number: OCE-1536521
2009-2013	National Centre for Earth Observation PhD studentship
2009	NERC subsidy to attend Earth System Science Spring School

2008-2009 NERC Masters studentship

Invited Talks

Talmy, D. Modeling microbial predator-prey interactions: From laboratory to the global ocean. Focus Session of the Center for the Mathematics of Biosystems, Virginia Tech, Virginia, USA, December 12th, 2025. Host: *Prof. Frank Aylward*.

Talmy, D. Modeling microbial predator-prey interactions: From laboratory to the global ocean. Ocean Ecology and Evolutionary Biology Seminar Series, University of Tennessee, Knoxville, USA, February 14th, 2025. Host: *Prof. Mike Gilchrist*.

Talmy, D. Empirical modeling of carbon flow through marine viruses and microzooplankton grazers. Ocean Carbon and Biogeochemistry workshop, Woods Hole Oceanographic Institution, Massachusetts, USA, June 11th, 2024. Hosts: *Prof. Sheri Floge, Prof. Jeff Bowman, and Prof. Jessica Labonte*.

Talmy, D. Modeling microbial host-virus interactions: from laboratory to global ocean. University of Bergen, Norway, June 15th, 2022. Host: *Dr. Selina Våge*.

Talmy, D. A model of algal-virus population dynamics reveals underlying controls on material transfer. Green Ocean workshop, École Normale Supérieure, Paris, France, May 18th, 2022. Host: *Prof. Corinne Le Quéré*.

Talmy, D. Nutrient enrichment and predation defense drive cyanobacteria population dynamics. University of Waterloo, Canada, March 11th, 2022. Host: *Prof. Jozef Nissimov*.

Talmy, D. Trade-offs between resource acquisition and defense shape cyanobacteria communities. University of Southern California, USA, October 13th, 2020. Host: *Prof. Naomi Levine*.

Talmy, D. Trade-offs modify ecosystem biomass structure along trophic gradients. SIMPLEX project kick-off meeting. University of Bergen, Norway, August 22nd, 2019. Host: *Prof. Selina Våge*.

Talmy, D. Inferring phytoplankton host-virus traits and trade-offs from laboratory population dynamics. Weizmann Institute of Science, Rehovot, Israel, August 14th, 2019. Host: *Prof. Assaf Vardi*.

Talmy, D. Understanding microbial ecosystem structure and function on large scales using mathematical models. American Society for Microbiology Annual Meeting, San Francisco, California, USA, June 21st, 2019. Host: *Prof. Alison Buchan*.

Talmy, D. Trait-based modeling of viruses in global ocean microbial ecosystems. Gordon Research Conference: Elucidating microbial processes across spatial and temporal scales. Lucca, Italy, July 5th, 2018. Host: *Prof. Kay Bidle*.

Talmy, D. What controls virus dynamics in global ocean microbial ecosystems? University of Texas, Austin, USA, March 23rd, 2018. Host: *Prof. Patrick Heimbach*.

Talmy, D. What controls microbial predator-prey ratios in the global ocean? Lamont-Doherty Earth Observatory, Columbia University, New York, USA, October 2nd, 2017. Host: *Prof. Andrew Juhl*.

Talmy, D., Martiny, A.C., Hickman, A.E., Follows, M.J. Microzooplankton regulation of particulate organic matter elemental composition. Bigelow Laboratory for Ocean Sciences, Maine, USA, March 9th, 2016. Host: *Dr. Nicolas Record*.

Talmy, D., Martiny, A.C., Hickman, A.E., Hill, C.N., Follows, M.J. Microzooplankton regulation of surface ocean POC:PON ratios. Rutgers University, New Jersey, USA, December 15th, 2015. Host: *Prof. Jozef Nissimov*.

Talmy, D., Martiny, A.C., Hickman, A.E., Follows, M.J. Zooplankton regulation of particulate organic matter elemental composition. National Oceanography Center, Southampton, UK, November 10th, 2015. Host: *Dr. Anna Hickman*.

Selected Oral Presentations

Underlined names indicate trainees in my research group

Hambrick, K.M., **Talmy, D.**, Wilhelm, S.W., Zinser, E.R. Temperature histories of host and phage influences infection of *Microcystis aeruginosa*. ASM KY/TN Branch Meeting, Knoxville, TN, April 11th, 2026.

Carr, E., Rajakaruna, H., Våge, S., Omta, A.W., **Talmy, D.** Killing the predator: impacts of top-predator mortality on global ocean ecosystem structure. Aquatic Sciences Meeting, Palma de Mallorca, Spain, June 9th, 2023.

McCullough, D.K., Calfee, B., Zinser, E.R., **Talmy, D.** Modeling the intersection of nutrient limitation and hydrogen peroxide on surface microbial communities. Aquatic Sciences Meeting, Palma de Mallorca, Spain, June 5th, 2023.

Papoulis, S.E., Wilhelm, S.W., **Talmy, D.**, Zinser, E.R. Nutrient enrichment and predation defense drive cyanobacteria population dynamics. Ocean Sciences Meeting, Virtual, March 3rd, 2022.

Rajakaruna, H., Omta, A.W., Carr, E., **Talmy, D.** The power-law biomass scaling relationship in natural microbial ecosystems. Ocean Sciences Meeting, Virtual, March 4th, 2022.

Talmy, D., Våge, S., Cael, B.B., Follows, M.J. Trade-offs modify ecosystem biomass structure along trophic gradients. 4th workshop on trait-based approaches to ocean life, Buckinghamshire, UK, August 18th, 2019.

Showalter, G.M., **Talmy, D.**, Deming, J.W. Modeling virus-host dynamics in sea ice brines. International Symposium on Sea Ice, Winnipeg, Manitoba, Canada, August 17-21st, 2019.

Hinson, A., **Talmy, D.** What controls algal-virus population dynamics in diverse systems? Aquatic Sciences Meeting, San Juan, Puerto Rico, February 26th, 2019.

Talmy, D., Bidle, K.D., Kranzler, C., Thamatrakoln, K. Inferring nutrient and light sensitivity of phytoplankton host-virus metabolism from laboratory population dynamics. 9th Aquatic Virus Workshop, Lincoln, Nebraska, USA, June 20th, 2018.

Talmy, D., Zakem, E., Follows, M.J. Modeling competitive interactions among viruses and microzooplankton grazers in the global ocean. Ocean Sciences Meeting, Portland, Oregon, USA, February 15th, 2018.

Talmy, D., Follett, C.L., Follows, M.J. Does viral lysis influence dissolved organic matter elemental composition? Aquatic Sciences Meeting, Honolulu, Hawaii, USA, February 28th, 2017.

Talmy, D., Thamatrakoln, K., Bidle, K.D., Follows, M.J. How does viral infection of *Emiliania huxleyi* depend on ambient irradiance? Monteiro group meeting, University of Bristol, Bristol, UK, July 26th, 2016.

Talmy, D., Hussain, F., Follows, M.J. The influence of lytic vs. lysogenic viral reproduction on host-virus populations. 8th Aquatic Virus Workshop, Plymouth, UK, July 11th, 2016.

Talmy, D., Martiny, A.C., Hickman, A.E., Hill, C.N., Follows, M.J. Microzooplankton regulation of surface ocean POC:PON ratios. Ocean Sciences Meeting, New Orleans, Louisiana, USA, February 25th, 2016.

Poster Presentations

Underlined names indicate trainees in my research group

Talmy, D., Carr, E. Modeling community dynamics: impacts of hydrogen peroxide on marine cyanobacteria composition. BioGeoScapes / C-CoMP Modeling Workshop, Woods Hole Oceanographic Institution, Massachusetts, USA, September 10th, 2025.

Talmy, D., Carr, E., Weitz, J., Dutkiewicz, S. Can viral induced mortality be represented implicitly and capture marine ecological dynamics? BioGeoScapes / C-CoMP Modeling Workshop, Woods Hole Oceanographic Institution, Massachusetts, USA, September 10th, 2025.

Talmy, D., Carr, E., Dutkiewicz, S., Weitz, J. Can viral induced mortality be represented implicitly and capture marine ecological dynamics? ASLO Aquatic Sciences Meeting, Charlotte, NC, USA, March 27th, 2025.

Hambrick, K.M., Smith, L.E., Martin, R.M., Stark, G.F., **Talmy, D.**, Wilhelm, S.W., Zinser, E.R. Temperature correlates with phage resistance in *Microcystis aeruginosa* NIES-298. ASLO Aquatic Sciences Meeting, Charlotte, NC, USA, March 27th, 2025.

Carr, E., Rajakaruna, H., **Talmy, D.** On the influence of mortality on microbial predator-prey relationships in the surface ocean. 5th workshop on trait-based approaches to ocean life, Knoxville, TN, January 27th, 2022.

McCullough, D.K., Calfee, B., Zinser, E.R., **Talmy, D.** Toward a general model of the effects of exogenous reactive oxygen on bacterial community dynamics. 5th workshop on trait-based approaches to ocean life, Knoxville, TN, January 27th, 2022.

Rajakaruna, H., Omta, A.W., Carr, E., **Talmy, D.** The power-law biomass scaling relationship in natural microbial ecosystems. 5th workshop on trait-based approaches to ocean life, Knoxville, TN, January 27th, 2022.

McCullough, D.K., Calfee, B., Zinser, E.R., **Talmy, D.** Connecting hydrogen peroxide damage and nutrient limitation controls on marine cyanobacterial growth. 4th workshop on trait-based approaches to ocean life, Buckinghamshire, UK, August 19th, 2019.

Papoulis, S.E., Wilhelm, S.W., **Talmy, D.**, Zinser, E.R. Environmental nutrients explain the distribution of restriction modification systems in prokaryotic genomes. 4th workshop on trait-based approaches to ocean life, Buckinghamshire, UK, August 20th, 2019.

Talmy, D., Follett, C.L., Follows, M.J. Viral regulation of dissolved organic matter elemental composition. Viruses of Microbes Meeting, Liverpool, UK, July 19th, 2016.

Workshops, Summer Schools, and Colloquia

2025	BioGeoScapes / C-CoMP Modeling Workshop, Woods Hole Oceanographic Institution, Massachusetts, USA.
2024	Ocean Carbon and Biogeochemistry Workshop, Woods Hole Oceanographic Institution, Massachusetts, USA.
2022	Microbes in Models: Steps for Integrating Microbial Activity into Climate Models, Virtual.
2022	Green Ocean Workshop, École Normale Supérieure, Paris, France.
2022	5th Trait-Based Approaches to Ocean Life Workshop, Knoxville, Tennessee, USA. Lead Organizer.
2020	NIMBioS Investigative Workshop on Quantitative Education in Life Sciences Graduate Programs, Virtual.

2019	4th Trait-Based Approaches to Ocean Life Workshop, Buckinghamshire, UK.
2018	9th Aquatic Virus Workshop, Lincoln, Nebraska, USA.
2016	8th Aquatic Virus Workshop, Plymouth, UK.
2015	2nd Trait-Based Approaches to Ocean Life Workshop, Waterville Valley, New Hampshire, USA.
2015	Workshop for Development of Macromolecular Models of Marine Algae, Sackville, New Brunswick, Canada.
2014	The Gordon and Betty Moore Foundation Research Associate and Postdoctoral Scholar Summit, Dorado, Puerto Rico.
2014	North Atlantic Virus Infection of Coccolithophores Expedition (Na-VICE) post cruise meeting, Woods Hole Oceanographic Institution, Falmouth, Massachusetts, USA.
2013	Trait and Resource Allocation Based Modeling of Microbial Communities, University of Exeter, UK.
2013	Workshop for Understanding Microbial Macromolecular Composition, Haifa, Israel.
2010	Advanced School on Complexity, Adaptation and Emergence in Marine Ecosystems, Trieste, Italy.
2010	Earth System Science Spring School, Scarborough, UK.
2010	Modeling Evolutionary and Ecological Processes in Biogeochemical Cycles, University of East Anglia, UK.

Teaching Experience

2026	Instructor, Reproducible Data Analysis (9 undergraduate and 5 graduate students), University of Tennessee, Knoxville.
2024-2025	Lecturer, General Microbiology (130 undergraduate students), University of Tennessee, Knoxville.
2023-2025	Instructor, Microbial Ecology Journal Club (20 graduate students), University of Tennessee, Knoxville.

2023	Instructor, Foundations in Microbiology (10 graduate students), University of Tennessee, Knoxville.
2022	Invited lecturer, Quantitative Methods in Virus Ecology and Evolution (10 graduate students), University of Bergen, Norway.
2020-2022	Lecturer, General Microbiology (100 undergraduate students), University of Tennessee, Knoxville.
2020-2021	Instructor, Advanced Topics in Microbiology (10 graduate students), University of Tennessee, Knoxville.
2018-2019	Instructor, Graduate Computational Biology Journal Club, University of Tennessee, Knoxville.
2016	Invited lecturer, Hjort Summer school in Bergen, Norway: Complexity vs. Simplicity in Microbial Ecology.
2008	Associate tutor, University of Sussex, UK. Courses taught: Linear algebra II and Introduction to Matlab.
2007	High school classroom assistant, Student Associate Scheme, Worthing High School, Worthing, East Sussex, UK.

Mentorship

Undergraduates

2024-2025	Francesca Ravelli, Math 498, Math Honors Thesis.
2023	Emily Landolt, NSF Research Experiences for Undergraduates, 10-week summer research placement.
2021	Kamryn Henderson, National Summer Undergraduate Research Project (NSURP), 8-week summer research project.
2020	Umang Joshi, Stephanie Westaway, Michael Lin, NIMBioS Research Experiences for Undergraduates, 6-week summer group research project.
2019	Lucas Fiet, Margie Knight, Priscilla Cho, NIMBioS Research Experiences for Undergraduates, 10-week summer group research project.

2019	Aaron Lin, NSF Research Experiences for Undergraduates, 10-week summer research placement.
2018-2019	Maitraya Ghatak, 401R, Undergraduate Research in Microbiology.
2018	Brielle Shortreed, NSF Research Experiences for Undergraduates, 10-week summer research placement.

Graduate Students

2025-present	Advisor and master's Committee Chair, Jackson Page-Roth, Microbiology Graduate Student, University of Tennessee. Master's thesis: *Jekyll and Hyde: Modeling bacterial pathogenicity modulation during algal blooms*.
2019-2021	Advisor for doctoral thesis chapter, Max Showalter, Astrobiology Program, University of Washington. Doctoral thesis chapter: *Modeling virus-host dynamics in sea ice brines*.
2018-present	Doctoral Committee member for 8 graduate students in the Department of Microbiology and the Department of Ecology and Evolutionary Biology, University of Tennessee.
2018-2024	Advisor and Doctoral Committee Chair, Katie McCullough, Microbiology Graduate Student, University of Tennessee. Doctoral thesis: *Modeling the impact of reactive oxygen species on ocean microbial community dynamics*.
2018-2022	Advisor and master's Committee Chair, Kyla Linn, Genome Science and Technology Graduate Student, University of Tennessee. Master's thesis: *Ecological controls on successional patterns in bloom forming cyanobacteria*.

Postdoctoral Researchers

2020-2022	Dr. Harshana Rajakaruna
2020-2021	Dr. Spiro Papoulis
2018-2020	Dr. Audra Hinson

Technical Professionals

2020-present	Eric Carr

Professional Service and Outreach

Service to the Discipline

Ongoing	Reviewer, *Science*, *Nature Communications*, *Nature Microbiology*, *mBio*, *Nature Ecology and Evolution*, *Science Advances*, *The ISME Journal*, *The American Naturalist*, *Limnology and Oceanography*, *Global Biogeochemical Cycles*, *Biological Invasions*, *Biophysical Journal*, *New Biotechnology*, *Journal of Theoretical Biology*, *Marine Ecology Progress Series*, *PLOS ONE*, *Frontiers in Marine Science*, *Journal of Marine Systems*
Ongoing	NSF Biological Oceanography, mail-in reviewer
2025-2027	Steering Committee member and organizer, Ocean Carbon and Biogeochemistry (OCB) workshop: "From single cells to ecosystems: quantifying the role of marine viruses in carbon transport models"
2025-present	Co-leader, Community of Scholars on Microbial Interactions and Communities
2023-2025	Steering Committee member, 7th Workshop on Trait-Based Approaches to Ocean Life
2022-2023	Steering Committee member, 6th Workshop on Trait-Based Approaches to Ocean Life
2022	Lead organizer, 5th Workshop on Trait-Based Approaches to Ocean Life, Knoxville, Tennessee
2021	NSF Panelist, Biological Oceanography Postdoctoral Fellowship program
2017	NASA panelist, EXPORTS program

Service to the Department and University

2026-present	University of Tennessee, Department of Microbiology, Faculty Mentoring Committee for Zach Burcham, committee member
2025-present	University of Tennessee, Department of Microbiology, Tenure Mentoring Committee for Jenny Heppert, committee member
2025-present	University of Tennessee, Department of Microbiology, Peer Teaching Committee for Sara Classen, committee member

2024-present	University of Tennessee, Department of Microbiology, SACSCOC assessment reporting for the Graduate Program, author
2024-2025	University of Tennessee, Department of Microbiology, Faculty Search Committee chair, Quantitative Microbe-Plant-Environment Interactions Assistant Professor search
2024-2025	University of Tennessee, Department of Microbiology, Faculty Search Committee member, Microbe-Animal Assistant Professor search
2023-2024	University of Tennessee, Division of Biology, Task force on beginning math courses for biology majors, member
2023-2024	University of Tennessee, Department of Microbiology, Faculty Search Committee member, Plant-Microbe Associate/Full Professor search
2020-2024	University of Tennessee, Department of Microbiology, Diversity, Equity, and Inclusivity Committee member
2020-2024	University of Tennessee, Department of Microbiology, member of the Committee for Curriculum and Undergraduate Affairs
2018-2020	University of Tennessee, Department of Microbiology, Graduate Admissions Committee member

Outreach and Community Engagement

2021-2022	Choice Health Network (LGBTQ+ community outreach and public health non-profit) Community Impact Committee Member
2018-2019	Annual Volunteer, Girl Scouts of Greater New York STEM programming events